

Low-resource Causal Relation Extraction With Semi-Supervised LLM Augmentation and Validation

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Abstract—Identifying causal relations in plain text is crucial for constructing knowledge graphs, as these relationships enhance our understanding of unstructured data. Traditional methods for causal relation extraction (CRE) often rely on large amounts of annotated data, which limits their effectiveness in real-world, data-scarce environments where noise and imbalance are prevalent. In this work, focusing on this problem, we propose a new approach that combines semi-supervised learning with large language models (LLMs) and incorporates validation to enhance the learning process. Particularly, our approach leverages LLMs to generate additional data and employs an iterative framework for dual validation. Evaluated across three datasets, our approach shows a significant 3% increase in F1 score. Additionally, in a low-resource setting using just 10% of the training data, we show that our approach achieves over a 5% improvement in F1 score compared to previous methods.

Causal relation extraction; data augmentation; large language models (LLMs); deep learning

I. INTRODUCTION

Causal relation extraction (CRE) is a crucial task in knowledge graph construction, concentrating on identifying causal relationships between entities or events within plain text [1,2]. For instance, in the sentence "The tsunami generates an earthquake", CRE seeks to identify the causal link between the events of the tsunami and the earthquake. The CRE task is essential for accurately building and enriching knowledge graphs with meaningful, interconnected information [3].

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Despite its importance, current studies in CRE face two major challenges: (1) Limited labeled data: Developing an effective CRE model requires a substantial amount of training data. Unfortunately, manually labeling this data is complex and time-consuming, leading to a scarcity of labeled examples. In specific domains, such as military and medicine, the costs associated with creating large datasets can be prohibitively high, which hampers the development and effectiveness of CRE methods. (2) Noise and chain characteristics: The limited availability of data can result in unstable and unreliable model training, leading to noisy predictions. Additionally, causal relationships often exhibit chain characteristics; for example, if event e_1 causes event e_2 , and event e_2 causes event e_3 , it logically follows that e_1 causes e_3 . Unfortunately, current methods rarely take these chain characteristics into account and tend to make independent predictions. These challenges have significantly impeded progress in CRE research.

In this paper, we address these two challenges by proposing a novel method that combines semi-supervised learning with Large Language Model (LLM) augmentation and a validation strategy. Specifically, to tackle the issue of limited labeled data, we introduce a semi-supervised learning framework designed to expand the training dataset. We propose three strategies to enhance the existing labeled data using prompt engineering, and we also suggest generating data from unlabeled text. By augmenting the training set with both labeled and unlabeled data, we demonstrate that our approach effectively increases the dataset size and improves the generalization capabilities of the CRE model.

Moreover, to mitigate noise and capture chain characteristics, we integrate a validation strategy enhanced by Large Language Models (LLMs) into the learning process. Specifically, we use LLMs to leverage their extensive knowledge and understanding of language to identify and filter out noise in the text while recognizing chain relationships between causal events. We also implement a validation step to ensure the accuracy and reliability of the extracted causal relations. Additionally, we introduce an iterative learning framework that facilitates a more fine-grained learning process, improving the performance of the CRE model and allowing for adjustments to model parameters, thereby minimizing noise and enhancing robustness.

To assess the effectiveness of our approach, we conducted extensive experiments across three diverse datasets. The results clearly demonstrate a significant improvement, with our method achieving a 5% increase in F1 score on standard datasets compared to baseline models. This improvement underscores the robustness of our approach in various contexts. Additionally, we evaluated our method in a low-resource scenario, which is often challenging for traditional models. In this setting, we observed substantial gains, including over a 10% increase in F1 score relative to the baseline models. These findings highlight the effectiveness of our approach in both typical and constrained environments, showcasing its potential for broader applications in causal relation extraction.

Our contributions are summarized as follows:

- (1) We propose a new approach for CRE, which combines semi-supervised learning with LLMs to address the challenges of limited training data, significantly reducing reliance on labeled datasets.
- (2) We introduce a high-confidence validation metric and an iterative framework that enhances the training process by effectively addressing noise and chain characteristics.
- (3) Our experiments across three datasets demonstrate promising results, including a 5% increase in F1 score and a 10% improvement in low-resource scenarios.

II. RELATED WORK

A. Causal Relation Extraction

Existing methods for causal relation extraction (CRE) can be classified into sentence-level and document-level approaches. Early sentence-level CRE methods primarily relied on lexical and syntactic patterns [3], causality cues or markers [4], and the joint recognition of temporal relations [5]. Recently, some approaches have incorporated external knowledge bases and weakly supervised methods to address issues of data scarcity and implicit causal relations [6]. More recent research has shifted towards integrating document-level features; for instance, Integer Linear Programming (ILP) has been utilized to comprehensively map global causal structures. Additionally, graph-based inference mechanisms have been effectively employed by models like DSGCN [7], which leverage Graph Convolutional Networks (GCNs) to represent relationships. Novel approaches, such as GIMC [8], utilize heterogeneous graph interaction networks to manage long-distance dependencies among causality spread across documents.

Despite these advancements, most CRE approaches struggle with limited data and the complexities of noise and chain characteristics, significantly hindering their applicability.

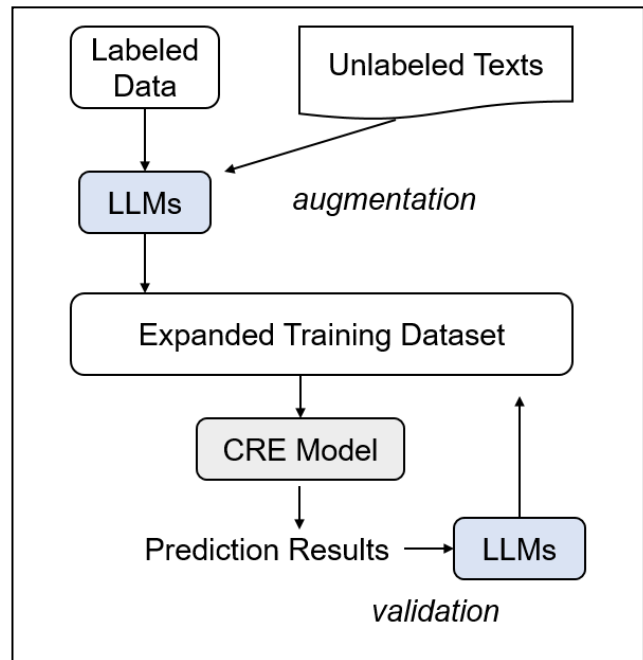


Figure 1. The overview of our approach.

B. Using LLMs for Data Augmentation

Using large language models (LLMs) for data augmentation involves generating diverse and high-quality text to enrich NLP training datasets. Techniques such as text generation, paraphrasing, and contextual expansion enable LLMs to produce coherent and semantically relevant variations of existing data [9]. This approach offers advantages like increased dataset diversity and improved semantic preservation, but it also presents challenges, including high computational demands and the potential for biased outputs. Recent research has explored multi-task learning and adversarial generation, highlighting the effectiveness of LLMs in creating robust training examples that enhance model performance across various NLP tasks [10]. However, to our knowledge, there remains a notable gap in the application of LLMs specifically to address the CRE task.

III. APPROACH

The overview of our approach is shown in Figure 1, which consists of three key components: (1) Semi-supervised LLM augmentation, which leverages both labeled data and unlabeled data to generate new training examples; (2) an LLM-based validation strategy for assessing the quality of predictions and enhancing learning; and (3) an iterative learning schema for continuous improvement. We will elaborate on each of these components in the sections below.

A. Semi-supervised LLM augmentation

In this component, we leverage large language models (LLMs) to perform data augmentation through a semi-supervised approach. Let D_l be the labeled dataset. We consider

both self-augmentation and augmentation from unlabeled data. For illustration, consider the labeled sentence: “The tsunami generates an earthquake,” where “tsunami” and “earthquake” express a causal relation.

In our self-augmentation, we propose three prompt engineering strategies aimed at modifying the sentence to create new training instances. These strategies include paraphrasing, generation, and fill-in-the-blank, as shown in Table 1.

- **Paraphrase.** We instruct the LLMs to rephrase the sentence without altering its semantics. For the original sentence, “The tsunami generates an earthquake,” we use the following prompt: “Please paraphrase the following sentence while maintaining its meaning: ‘The tsunami generates an earthquake.’”. The LLM could respond with, “The tsunami causes an earthquake”.
- **Generation.** In this strategy, we retain the two main events from the original sentence and ask the LLMs to generate a new sentence that maintains the same meaning. For example, we might prompt with, “Generate a new sentence that includes the same events as this one: ‘The tsunami generates an earthquake.’” A possible response could be, “An earthquake is triggered by the tsunami”.
- **Fill-in-the-blank.** Similar to BERT, we randomly remove words from the sentence and use LLMs to generate a new example that conveys the same events. We provide a complete instructional prompt as shown in Table 1.

While the self-augmentation strategies described above provide valuable augmentations, they may generate limited variations from existing labeled data. To address these limitations, we propose a new augmentation method using an external dataset, denoted as D_u for the unlabeled dataset. This method involves two steps:

- **Identifying Potential Causal Relations:** The first step is to find sentences that express causal relations. We utilize LLMs for this task, employing the prompt: “Does the sentence demonstrate a causal relation?”
- **Identifying Two Entities/Events:** In this step, we focus on identifying two entities or events necessary for model training. We again use LLMs for this identification, considering only sentences that contain exactly two entities or events as valid training data.

By employing these strategies, we can significantly expand the training set for model learning.

B. Model learning with LLM-based validation

Building on previous approaches, we develop a BERT-based model with binary classification for causal relation extraction. This model first uses BERT to encode the sentence into vector representations and then constructs a basic binary classification model to predict whether a causal relationship exists between two entities or events.

Let the sentence be denoted as S , with the two entities/events as e_1 and e_2 . The BERT encoding process can be expressed as:

$$H = \text{BERT}(S) \quad (1)$$

TABLE I. PROMPTS FOR SELF-AUGMENTATION

Strategy	Prompt
Paraphrase	Please paraphrase the following sentence while maintaining its meaning: ‘The tsunami generates an earthquake..’
Generation	Generate a new sentence that includes the same events as this one: ‘The tsunami generates an earthquake.’
Fill-in-the-blank	Complete the following sentence by filling in the blanks: ‘The _____ generates an _____.’

where BERT denotes the BERT encoder, and H is the representation matrix for the sentence. We further define the binary classification model as follows:

$$c = \sigma(W * (H \oplus h_1 \oplus h_2)) \quad (2)$$

In this formulation, W is a parameter matrix, h_1 and h_2 are the representations of e_1 and e_2 , and σ denotes the sigmoid function.

Nevertheless, due to limited training data and potential noise in the augmented data, predictions may become unstable. To address this, we propose two strategies for using large language models (LLMs) as validation techniques to enhance learning.

The first strategy involves handling predictions with probabilities in the range of (0.5, 0.7]. This range indicates uncertainty in the causal relation extraction (CRE) model. In such cases, we use LLMs to generate explanations for the entities or events involved and append this explanation to the original sentence for re-prediction. This process is repeated until the model reaches a confident prediction.

Our second validation strategy focuses on ensuring global consistency in predictions. Specifically, we utilize LLMs to verify that if our model predicts e_1 causes e_2 and e_2 causes e_3 , it should also predict that e_1 causes e_3 . If the model fails to make this transitive prediction, it may indicate an error. In such cases, we generate explanations using LLMs and concatenate them with the original sentence, similar to the first strategy, until a consistent prediction is achieved. By enforcing consistent predictions, we aim to improve the reliability of our model's outputs.

C. An iterative learning schema

We further integrate the above strategies into the training process using an iterative learning schema. The process is as follows:

- For each training example, after tuning the model, we use the trained model to re-predict the label for the two entities or events. If the model still generates a less confident prediction (i.e., within the range of (0.5, 0.7]), we employ LLMs to generate additional clues and concatenate these clues with the original sentence for training. This approach helps to create more informative examples for learning.

- The second aspect focuses on chained reasoning. The criteria are similar: for each training example, we first generate predictions for the relationships between each pair of entities or events. If any pair does not align with global consistency, we utilize LLMs to generate further clues for learning until global consistency is achieved.

This process is performed for each training example and for all event pairs. By doing so, we incorporate more necessary clues into the learning process, enhancing the model's performance.

IV. EXPERIMENTAL SETTINGS

We show the detailed experimental settings in this section.

A. Datasets and evaluations

We utilize three existing datasets for causal relation extraction: CNC, CTB, and PDTB-2. Due to the unavailability of publicly accessible labels for the CNC test set, we use the CNC validation set as our test set. To simulate a low-resource setting, we split the CNC training set into ten parts and randomly select 10% of the data for model training in each iteration. This approach allows us to evaluate the effectiveness of our method in a low-resource environment. Our evaluation metrics include precision (P), recall (R), and F1 score.

B. Baselines

We compare our approach against three baseline methods: 1) LLMs, which directly utilize large language models for causal relation extraction, specifically using LLAMA 3 as the foundational model; 2) BERT, which employs BERT to generate representations for prediction; and 3) KG-CRE, which enhances the input by incorporating knowledge graphs.

V. RESULTS AND DISCUSSION

A. Main Results

Table 2 presents the results for the various methods evaluated. Notably, our approach demonstrates strong performance, outperforming the baselines by at least 3% in absolute F1 score. This improvement is particularly pronounced when compared to the BERT model, which does not utilize data augmentation.

Upon examining the performance of previous methods, we observe that LLMs yield relatively poor results. This can be attributed to their limitations in making indicative predictions, which affects their overall effectiveness in causal relation extraction. In contrast, methods that incorporate knowledge graph (KG) augmentation achieve better performance, as they leverage additional context about entities and events.

By directly comparing our approach to BERT, we highlight the significant advantages provided by data augmentation and validation. This not only enhances the model's predictive capabilities but also underscores the importance of incorporating external knowledge in improving causal relation extraction.

B. Results in the low-resource setting

Table 3 compares the performance of different methods in a low-resource setting, using CNC as the testbed. Notably, our approach demonstrates significantly better performance than previous methods, achieving an absolute improvement of over 5%. This outcome is understandable given the constraints of a

TABLE II. RESULTS ON DIFFERENT DATASETS

Datasets	Model	P	R	F1
CNC	LLAMA 3	81.1	80.5	80.8
	BERT	83.2	80.5	81.8
	KG-CRE	83.6	83.1	83.4
	Ours	85.4	88.5	86.9
CTB	LLAMA 3	83.9	34.6	49.0
	BERT	84.3	34.6	49.1
	KG-CRE	85.5	36.5	51.2
	Ours	87.2	39.5	54.4
PDTB-2	LLAMA 3	73.9	85.6	79.3
	BERT	75.9	85.2	80.3
	KG-CRE	78.4	85.4	81.7
	Ours	83.4	87.5	85.3

TABLE III. RESULTS IN THE LOW-RESOURCE SETTING

Model	P	R	F1
LLAMA 3	58.0	52.4	55.1
BERT	58.2	54.4	56.2
KG-CRE	59.1	58.4	58.7
Ours	64.4	65.5	64.9

limited training set, which means that the models' effectiveness increasingly depends on external resources.

C. Ablations

We conducted an ablation study on different components of our model, specifically focusing on the full model's performance on the CNC dataset. The results reveal the following: removing data augmentation results in a performance drop of 2.3%; excluding data augmentation from unlabeled data leads to a decrease of 2.1% in F1 score; and omitting the validation component causes a drop of 2.8%. These findings suggest that all components of the model are crucial for effective learning. In particular, leveraging existing labeled data appears to be more beneficial than generating information solely from unlabeled data. Additionally, the validation process plays an important role in addressing noise and ensuring global consistency within the model's predictions.

VI. CONCLUSIONS

In this work, we address the challenges of causal relation extraction (CRE). We propose a novel approach that integrates semi-supervised learning with large language models (LLMs) and incorporates validation to enhance the learning process. Experimental results show a 3% increase in F1 score of our approach on standard datasets, and in a low-resource setting with

just 10% of the training data, our approach achieves over a 5% improvement compared to previous methods

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